# Experiment - 2

Aim:- ER Diagram to Relational Schema

1. Relational Schema

| Table | Primary Key | Foreign Keys |
|---|---|---|
| CUSTOMER | Customer ID | — |
| CUSTOMER_PHONE | (Customer ID, Phone) | CustomerID → CUSTOMER |
| CUSTOMER_ALT_PHONE | (CustomerID, AltPhone) | CustomerID → CUSTOMER |
| SELLER | SellerID | — |
| SELLER_PHONE | (Seller ID, Phone) | SellerID → SELLER |
| ADDRESS | AddressID | Customer ID → CUSTOMER |
| ADDRESS_MOBILE | (AddressID, MobileNo) | AddressID → ADDRESS |
| CATEGORY | CategyID | Parent CategoryID → CATEGORY |
| PRODUCT | ProductID | SellerID → SELLER, CategoryID → CATEGORY |
| PRODUCT_IMAGE | (ProductID, ImageURL) | ProductID → PRODUCT |
| ELECTRONICS | ProductID | ProductID → PRODUCT |
| CLOTHING | ProductID | ProductID → PRODUCT |
| BOOKS | ProductID | ProductID → PRODUCT |
| ORDERS | OrderID | CustomerID → CUSTOMER, AddressID → ADDESS |
| ORDER_COUPON | (OrderID, Coupon Code) | OrderID → ORDERS |
| ORDERITEM | (OrderID, ItemNo) | OrderID → ORERS, ProductID → PRODUCT |
| PAYMENT | PaymentID | OrderID → ORDERS |
| DELIVERY | DeliveryID | OrderID → ORDERS, AddressID → ADDRESS |

```
CREATE DATABASE ECOMERCE;
USE ECOMMERCE;
CREATE TABLE CUSTOMER(
CustomerID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Email VARCHAR(100) NOT NULL UNIQUE,
Password VARCHAR(100) NOT NULL,
DateOfBirth Date NOT NULL
);
CREATE TABLE CUSTOMER_PHONE(
CustomerID INT,
Phone VARCHAR(15),
PRIMARY KEY (CustomerID, Phone)
FOREIGN KEY(CustomerID) REFERENCES CUSTOMER (CustomerID)
ON DELETE CASCADE
);
CREATE TABLE CUSTOMER_ALT_PHONE (
CustomerID INT,
AltPhone VARCHAR(15),
PRIMARY KEY (CustomerID, AltPhone),
FOREIGN KEY (CustomerID) REFERENCES CUSTOMER (customerID)
ON DELETE CASCADE
);
CREATE TABLE SELLER(
SellerID INT PRIMARY KEY,
SellerName VARCHAR(50) NOT NULL ,
Email VARCHAR(100) UNIQUE NOT NULL,
GSTIN VARCHAR (20) UNIQUE NOT NULL,
AccountNo VARCHAR(20),
IFSCCode VARCHAR (20),
BANKName VARCHAR (50),
);
CREATE TABLE SELLER_PHONE(
SellerID INT,
Phone VARCHAR(15),
PRIMARY KEY (SellerID, Phone),
FOREIGN KEY (SellerID) REFERENCES SELLER (SellerID)
ON DELETE CASCADE
);
```

```
CREATE TABLE CATEGORY (
CategoryID INT PRIMARY KEY,
CategoryName VARCHAR(50) NOT NULL,
Description VARCHAR(100),
ParentCategoryID INT,
FOREIGN KEY (ParentCategoryID) REFERENCES CATEGORY(CategoryID)
ON DELETE SET NULL
);

CREATE TABLE PRODUCT (
ProductID INT PRIMARY KEY,
ProductName VARCHAR(50) NOT NULL,
Description VARCHAR(100),
Price DECIMAL(10,2) NOT NULL,
StockQty INT NOT NULL,
Weight DECIMAL(8,2),
SellerID INT NOT NULL,
CategoryID INT,
FOREIGN KEY(sellerID) REFERENCES SELLER(sellerID)
ON DELETE CASCADE,
FOREIGN KEY (CategoryID) REFERENCES CATEGORY(CategoryID)
ON DELETE SET NULL
);

CREATE TABLE PRODUCT_IMAGE (
ProductID INT,
Image VARCHAR(200),
PRIMARY KEY(ProductID, Image),
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
ON DELETE CASCADE
);

CREATE TABLE ELECTRONICS(
ProductID INT PRIMARY KEY,
Brand VARCHAR(50),
Warranty VARCHAR(30),
PowerRating VARCHAR(30),
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
ON DELETE CASCADE
);
```

```
CREATE TABLE CLOTHING (
ProductID INT PRIMARY KEY,
 Size VARCHAR (20),
 Colour VARCHAR (20),
 Material VARCHAR (30),
 FOREIGN KEY (ProductID) REFERENCES PRODUCT (ProductID)
 ON DELETE CASCADE
 );

CREATE TABLE BOOKS (
ProductID INT PRIMARY KEY,
ISBN VARCHAR (20) UNIQUE,
Author VARCHAR (50),
Publisher VARCHAR (50),
FOREIGN KEY (ProductID) REFERENCES PRODUCT (ProductID)
ON DELETE CASCADE
 );

CREATE TABLE ADDERSS (
AddressID INT PRIMARY KEY,
DoorNo VARCHAR (10),
 Street VARCHAR (50),
 Area VARCHAR (50),
 City VARCHAR (30),
 State VARCHAR (30),
 Pincode VARCHAR (10) NOT NULL,
 AddressType VARCHAR (20),
 CustomerID INT NOT NULL,
 FOREIGN KEY (CustomerID) REFERENCES CUSTOMER (CustomerID)
 ON DELETE CASCADE
 );

CREATE TABLE ADDRESS_MOBILE (
AddressID INT,
MobileNo VARCHAR (15),
PRIMARY KEY (AddressID, MobileNo),
FOREIGN KEY (AddressID) REFERENCES ADDESS (AddressID)
ON DELETE CASCADE
 );
```

```
CREATE TABLE ORDERS (
OrderID INT PRIMARY KEY,
OrderDate DATE NOT NULL,
OrderStatus VARCHAR(20) NOT NULL,
TotalAmount DECIMAL(10,2) NOT NULL,
CustomerID INT NOT NULL,
AddressID INT,
FOREIGN KEY (CustomerID) REFERENCES CUSTOMER (CustomerID)
ON DELETE CASCADE,
FOREIGN KEY (AddressID) REFERENCES ADDRESS (AddressID)
ON DELETE SET NULL.
);

CREATE TABLE ORDER_COUPON (
OrderID INT,
CouponCode VARCHAR(30),
PRIMARY KEY (OrderID, CouponCode),
FOREIGN KEY (OrderID) REFERENCES ORDERS (orderID)
ON DELETE CASCADE.
);
CREATE TABLE ORDERITEM (
OrderID INT,
ItemNo INT,
ProductID INT NOT NULL ,
ItemPrice DECIMAL(10,2) NOT NULL,
Quatity INT NOT NULL,
Discount Decimal(10,2),
Tax Amount DECIMAL(10,2),
PRIMARY KEY (OrderID, ItemNo),
FOREIGN KEY (OrderID) RERERECES ORDERS (orderID)
ON DELETE CASCADE,
FOREIGN KEY (ProductID) REFERENCES PRODUCT (ProductID)
ON DELETE CASCADE
);

CREATE TABLE PAYMENT (
PaymentID INT PRIMARY KEY,
PaymentDate DATE NOT NULL,
Amount DECIMAL(10,2) NOT NULL,
```

```
PaymentMethod VARCHAR(30) NOT NULL,
TransactionID VARCHAR(30) UNIQUE,
PaymentStatus VARCHAR(20) NOT NULL,
OrderID INT UNIQUE NOT NULL,
FOREIGN KEY (orderID) REFERENCES ORDERS (orderID)
ON DELETE CASCADE
);
CREATE TABLE DELIVERY (
DeliveryID INT PRIMARY KEY,
DeliveryDate DATE,
DeliveryStatus VARCHAR(20) NOT NULL,
TrackingID VARCHAR(30) UNIQUE,
DeliveryPartner VARCHAR(50),
ShippingCharge DECIMAL(8,2),
OrderID INT UNIQUE NOT NULL,
AddressID INT,
FOREIGN KEY (orderID) REFERENCES ORDERS (orderID)
ON DELETE CASCADE,
FOREIGN KEY (AddressID) REFERENCES ADDRESS (AddressID)
ON DELETE CASCADE,
);

INSERT INTO CUSTOMER VALUES
(1, 'Priyanshu', 'priyanshumishra02007@gmail.com', 'pass123', '2007-01-20');

INSERT INTO CUSTOMER_PHONE VALUES
(1, '6388046065');

INSERT INTO CUSTOMER_ALT_PHONE VALUES
(1, '6388046065');

INSERT INTO SELLER VALUES
(101, 'TechStore', 'tech@gmail.com', 'GST123',
'123456', 'SBIN0002', 'SBI');

INSERT INTO SELLER_PHONE_VALUES
(101, '7897897919');
```

```
INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic items', NULL);

INSERT INTO CATEGORY VALUES
(202, 'Mobiles', 'Mobile phones', 201);

INSERT INTO PRODUCT VALUES
(301, 'Laptop', 'Gaming Laptop', 75000, 10, 2.5, 101, 201);

INSERT INTO ELECTRONICS VALUES
(301, 'ASUS', '2 Years', '180W');

INSERT INTO PRODUCT_IMAGE VALUES
(301, 'laptop.jpg');

INSERT INTO ADDRESS VALUES
(401, '12', 'Nawabganj', 'Unnao', 'Uttar Pradesh', '209859', 'Work', 1);

INSERT INTO ADDRESS_MOBILE VALUES
(401, '919295,8085');

INSERT INTO ORDERS VALUES
(501, '2026-08-14', 'Placed', 75000, 1, 401);

INSERT INTO ORDERITEM VALUES
(501, 1, 301, 95000, 1, 1000, 9000);

INSERT INTO ORDER_COUPON VALUES
(501, 'WELCOME10');

INSERT INTO PAYMENT VALUES
(601, '2026-08-14', 74000, 'UPI', 'TXN123', 'Paid', 501);

INSERT INTO DELIVERY VALUES
(701, NULL, 'Processing', 'TRK123', 'Delhivery', 100, 501, 401);

-- RERERENTIAL INTEGRITY VIOLATION {INSERT INTO ORDERS VALUES
(502, (2026-08-14', 'Placed', 50000, 999, 401);

-- SET NULL DEMONSTRATION
DELETE FROM CATEGORY WHERE CategoryID = 201;

-- CASCADE DEMONSTRATION
DELETE FROM CUSTOMER WHERE CustomerID = 1;
```

## OUTPUT 1 — SHOW TABLES

After creating all tables:

SHOW TABLES;

| Tables_In_ecommerce |
|---|
| ADDRESS |
| ADDRESSS_MOBILE |
| BOOKS |
| CATEGORY |
| CLOTHING |
| CUSTOMER |
| CUSTOMER_ALT_PHONE |
| CUSTOME_PHONE |
| DELIVERY |
| ELECTRONICS |
| ORDERS |
| ORDERITEM |
| ORDER_COUPON |
| PAYMENT |
| PRODUCT |
| PRODUCT_IMAGE |
| SELLER |
| SELLER_PHONE |

18 rows

It proves our relational schema

## OUTPUT 2 — Sample data

Use:

```
SELECT * FROM CUSTOMER;
SELECT * FROM PRODUCT;
SELECT * FROM ORDERS;
```

CUSTOMER

| CustomerID | Name | Email | Password | Date of Birth |
|---|---|---|---|---|
| 1 | Priyanka | priyankumari[illegible]@gmail.com | Pass123 | 2007-01-20 |

PRODUCT

| ProductID | ProductName | Description | Price | StockQty | Weight | SellerID | CategoryID |
|---|---|---|---|---|---|---|---|
| 301 | Laptop | Gaming Laptop | 75000.00 | 10 | 2.50 | 101 | 201 |

ORDERS

| OrderID | OrderDate | OrderStatus | TotalAmount | CustomerID | AddressID |
|---|---|---|---|---|---|
| 501 | 2026-08-14 | Placed | 75000,00 | 1 | 401 |

Proves sample data insertion

## OUTPUT 3 — Referential integrity violation

```
INSERT INTO ORDERS VALUES
(502,'2026-08-14','Placed',50000,999,401);
```

Since CustomerID = 999 doesn't exist:

```
ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails
```

Reason: CustomerID 999 does not exist in CUSTOMER.

Satisfies "demonstrate referential integrity violations."

## OUTPUT 4 — ON DELETE SET NULL

AIM explicitly asks us to use SET NULL.

Before deletion:

```
SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;
```

Draw:

| ProductID | ProductName | CategoryID |
|---|---|---|
| 301 | Laptop | 201 |

Then:

```
DELETE FROM CATEGORY WHERE CategoryID = 201;
```

Then:

```
SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;
```

Draw:

| ProductID | ProductName | CategoryID |
|---|---|---|
| 301 | Laptop | NULL |

Write underneath:

ON DELETE SET NULL demonstrated successfully.

## OUTPUT-5 — ON DELETE CASCADE

Before deleting:

```
SELECT * FROM ORDERS;
```

Output:

| OrderID | OrderDate | OrderStatus | TotalAmount | CustomerID | AddressID |
|---|---|---|---|---|---|
| 501 | 2026-08-14 | Placed | 75000.00 | 1 | 401 |

Then:

```
DELETE FROM CUSTOMER WHERE CustomerID = 1;
```

Output:

Query OK, 1 row affected

Then:

```
SELECT * FROM ORDERS;
```

Output:

Empty set

You can also show:

```
SELECT * FROM CUSTOMER_PHONE;
```

Output:

```
Empty set
```

This demonstrates that deleting the parent customer automatically deleted the related records.

Write:

ON DELETE CASCADE demonstrated successfully.

